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A multi-task-type bankruptcy prediction method without using financial statements

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Abstract

We propose a new bankruptcy prediction method using features that do not depend on financial statements. Furthermore, we show that multi-task learning effectively constructs a prediction model with small-sized samples. In conventional bankruptcy prediction, corporate features derived from balance sheets and profit/loss statements are employed as inputs for prediction. Thus, predicting bankruptcies of firms that do not prepare financial statements has been difficult. To realize default prediction without financial statements, we propose the use of transaction information such as the credit-worthiness of the business counterparty and the dispersion of business relationships. In addition, we verify the availability of "distance to default" which is a feature derived from the structural approach of credit risk assessment. To cope with small sample size situations, we apply multi-task learning for prediction model construction. Empirical results on Japanese corporate samples show that some of the proposed features include useful information for bankruptcy prediction. Furthermore, we confirmed that the prediction accuracy of multi-task learning outperforms that of conventional single-task learning in some cases.

Keywords Credit risk · Bankruptcy · Multi-task learning · Least-squares probabilistic classifier

Mathematics Subject Classification 91-10 · 91G40

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1 Introduction

Corporate bankruptcy prediction has long been a significant issue in quantitative financial risk management, and it has attracted the attention of researchers and practitioners [1]. Most conventional bankruptcy prediction methods are based on features derived from financial statements. For this reason, conventional prediction methods are difficult to apply to companies that do not prepare financial statements. Therefore, in this study, we investigate the feasibility of using the new features derived from the transaction information obtained from company surveys, i.e., interviews with companies, for bankruptcy forecasting. Furthermore, another difficulty in developing bankruptcy prediction methods is that sometimes the sample size of available data for developing prediction models is too small. To address that small-sample size problem, we will demonstrate the use of multi-task learning as a model construction method, and confirm if it is possible to construct the model even when the size of samples is small.

Traditionally, financial index values that have been generated from financial statements of companies such as balance sheets and profit/loss statements are employed as input values in default prediction models or credit scoring models. The credit score is a creditworthiness indicator that expresses the probability that a firm will default or go bankrupt. Over the years, model validation has confirmed that the accuracy of bankruptcy projections based on financial statements is sufficient for practical use. It can be said that default prediction is possible when at least several periods of financial statements of the target company are available. However, many relatively small firms do not prepare financial statements, then it is difficult to apply a bankruptcy prediction model that uses only features derived from financial statements as inputs.

Based on the above-mentioned issues, this study aims to find a method that enables bankruptcy prediction even for firms without financial statements. Even if it is difficult to obtain financial statements, it is possible to observe sales and net income through surveys conducted by a corporate research organization. In addition, information on the business transaction such as a list of business counterparts is observable through corporate surveys. Moreover, comprehensive corporate surveys conducted by private research organizations make it possible to determine the share of the value of transactions between the target firm and its main counterparts. Since the business counterparts of the target firm are often long-established companies or large companies, the financial statements of the business counterparts are observable and the credit scores of the counterparts are calculated using their financial statements. That is, we are able to use information on the degree of counterparty diversification and the creditworthiness of the counterparties. Accordingly, we employ the features on credit scores of counterparts, and futures on transaction diversifications for our default prediction.

Even when whole financial statements are not available, we can partially obtain financial information such as sales and debt for several years by company surveys. To use the information on observed time-series data of sales and debts, we focus on the structural approach to credit risk assessment. The structural approach calculates the probability of bankruptcy using a stochastic process model that explicitly expresses the structure of bankruptcy. We propose a new set of features reflecting the level and

volatility of sales, derived by combining the profit-based structural approach [2] with a simple model representing the relationship between sales and profits.

In general, a large sample of firms that can be considered homogeneous except for differences in features is required as data for building a bankruptcy prediction model. Specifically, practitioners construct industry-specific bankruptcy prediction models using a sample consisting of firms in the same industry. However, there are cases in which homogeneous firm data with a sufficient sample size cannot be obtained. Furthermore, in general, a corporate database is an imbalanced dataset, in which the number of observed bankrupt firms and the number of non-bankrupt firms are very different. The sample size for model construction is significantly smaller when under-sampling is used, which is known as the standard way to deal with unbalanced data. Therefore, there is a need for a method that allows models to be built even in situations where the sample size is extremely small.

To deal with the small sample size situations, we attempt to build bankruptcy prediction models by multi-task learning. Multi-task learning is a method to improve the generalization performance of a model by learning multiple tasks simultaneously. This learning method is known to have the same effect as increasing the number of training data samples. In this study, we demonstrate that it is possible to construct a model for categories with a small samples by multi-task learning. As a specific forecasting method, we employ multi-task least-squares probablistic classifier (MT-LSPC) proposed in [3]. MT-LSPC is an extension of least-squares probablistic classifier (LSPC) [4] to multi-task learning. The efficiency of LSPC for credit scoring using financial statements is shown in [5].

The remainder of this paper is organized as follows: Sect. 2 briefly describes the formulation of the default prediction problem. Section 3 introduces new firm features without using financial statements. Section 4 describes the formulation of LSPC and the MT-LSPC. Section 5 provides empirical analysis. Section 6 highlights the conclusions and directions for future research.

2 Bankruptcy prediction as probabilistic classification problem

Bankruptcy prediction is a two-class classification problem. We suppose that our samples for bankruptcy prediction are $\{(\boldsymbol{x}_i,\ y_i,\ t_i)\}_{i=1}^N$. Here, the label $y_i \in \{0,\ 1\}$ represent the non-default ($y_i = 0$) or default ($y_i = 1$) for sample firm i. The vector $\boldsymbol{x}_i = (x_{i1},\ x_{i2}, \ldots, x_{ip})$ represents the sample-specific features associated with the firm i. Index $t_i \in \{1, 2, \ldots, T\}$ represents the sector which the firm i belongs to. Default prediction aims to determine the state of a firm at a particular time, given the values of the features. For this purpose, we employ probabilistic classification. We estimate class-posterior probability $p(y|\boldsymbol{x}, t)$ from the samples $\{(\boldsymbol{x}_i, y_i, t_i)\}_{i=1}^N$, then classify the sample $\boldsymbol{x}$ to class $\hat{y}$ by

$$\hat{y} := \mathrm{argmax}_y\, p(y|\boldsymbol{x}, t). \tag{1}$$

3 Corporate features for bankruptcy prediction

In this section, we specify the corporate features $\boldsymbol{x}_i = (x_{i1}, x_{i2}, \ldots, x_{ip})$ for our default prediction. In order to include companies for which financial statements are not available, we propose a feature set that is derived from business transaction information. That is, we generate features derived from information on sales and payments to major customers and suppliers, as well as information on the credit scores of customers and suppliers. The original data for generating the features can be obtained through company surveys conducted by credit reporting agencies. In addition, we propose a set of features derived from a structural-type credit risk assessment.

3.1 Corporate features on business transactions

Here, we describe new features for bankruptcy prediction, which are calculated from information on business transactions. To derive our features, we prepare the payments from firm j to firm i, which we denote S_{ji} $(i, j = 1, 2, \ldots, N)$. Here, we set $S_{i,i} = 0$. Then, we obtain the sales share of customer j on total sales of target firm i by $s_{ji} = S_{ji} / \sum_k S_{ki}$. Similarly, we obtain the payments share of main supplier j on total payments of target firm i by $s_{ij} = S_{ij} / \sum_k S_{ik}$.

The first feature for describing transaction status of firm i, which we call credit score weighted transaction shares (CTS), is as follows:

$$CTS_i^{in} = \sum_{j=1}^{J} s_{ji} c_j \tag{2}$$

Here, c_j represents the credit score of customer j, and we suppose that firms $j = 1, 2, \ldots$ are ordered in descending order of transaction share s_{ji}. Thus, CTS_i^{in} represents the weighted average of creditworthiness of customers with top J share for target firm i. We also introduce a weighted average of the creditworthiness of supplier that sells products or services to target firm i, as follows:

$$CTS_i^{out} = \sum_{j=1}^{\tilde{J}} s_{ij} c_j \tag{3}$$

Here, we suppose that firms $j = 1, 2, \ldots$ are ordered in descending order of transaction share s_{ij}. Thus, CTS_i^{out} represents the weighted average of creditworthiness of suppliers with top J share for target firm i.

Some measures for transaction diversification are the candidate for $\boldsymbol{x}_i = (x_{i1}, x_{i2}, \ldots, x_{ip})$. Especially, we employ the well-known Herfindahl–Hirschman Index (HHI) for measuring the diversification of sales shares on customers:

$$HHI_i^{in} = \sum_{j=1}^{N} s_{ji}^2 \tag{4}$$

The HHI is a measure of share concentration, often used to assess the level of competition power of a firm within an industry or sector. It quantifies the concentration of the share by considering the size and number of firms. When the value of HHI is near to 0, there are many small firms with equal shares. When the value of HHI is 1, one firm controls the entire market. We also consider HHI on pay-side diversification:

$$HHI_i^{out} = \sum_{j=1}^{N} s_{ij}^2 \tag{5}$$

3.2 Corporate features derived from the structure of bankruptcy occurrence

In addition to the above new features, we propose another feature derived from a rather simple credit risk model within a structural approach. The structural approach for credit risk assessment is a method that explicitly captures the financial condition corresponding to corporate default [1]. The idea of the structural approach can be stated simply as follows. Denote the underlying filtered complete probability space by $(\Omega,\ \mathcal{F},\ \mathbb{P},\ \{\mathcal{F}_t\}_{t\geq 0})$, where $\{\mathcal{F}_t\}_{t\geq 0}$ is a right-continuous and complete filtration. The probability measure $\mathbb{P}$ is assumed to represent the physical measure Let V_t be the value of corporate assets at time t and D be the liabilities, and let $\{V_T < D\}$ be the event of insolvency at a future time T. The probability of insolvency $P(V_T < D)$ is calculated with a stochastic process that represents the time-series transition of corporate asset value V_t and debt amount D.

One of the credit risk metrics derived from the structural approach is the "Distance to Default" (DD) [6]. A typical model of the structural approach is one in which the time transition of corporate assets is modeled by geometric Brownian motion $dV_t = \mu^V V_t dt + \sigma^V V_t dW_t^V$, then the value of the corporate asset at time T follows a log-normal distribution. In this case, the probability of an event $V_T < D$, which corresponds to a default condition of insolvency, is

$$P(V_T < D) = \Phi\left(-\frac{\log V_0 - \log D + \left(\mu^V - \frac{1}{2}(\sigma^V)^2\right)T}{\sigma^V\sqrt{T}}\right).$$

Here, function $\Phi(\cdot)$ the cumulative distribution function of the standard normal distribution. Let us focus on the expression for the threshold value:

$$DD_V = \frac{\log V_0 - \log D + \left(\mu^V - \frac{1}{2}(\sigma^V)^2\right)T}{\sigma^V\sqrt{T}}.$$

When DD_V is larger, the probability of insolvency is smaller. Therefore, DD_V is called Distance-to-Default, and the metrics are one of the well-known indicators representing the magnitude of credit risk.

As noted above, in the derivation of the traditional default distance DD_V, we directly assumed a random variable that represents the value of corporate assets. Our objective is to obtain a credit evaluation index derived from information on sales.

Then, let us consider the DD derived from stochastic models of sales. According to the finance theory, the value of corporate assets is generally given as the discounted present value of future cash flows. Therefore, if the instantaneous corporate profit at time t is denoted by η_t, the value of corporate assets at time T is given by

$$V_T^H = E_T\left[\int_T^H \exp(-\rho t)\eta_t dt\right]$$

Here, $\rho > 0$ denotes the risk-adjusted discount rate, and H denotes the time horizon at which an asset ceases to generate cash flow. Assuming an appropriate profit fluctuation model, the probability distribution of corporate asset values can be derived under the model, and the probability of insolvency can be calculated. A simple model of profit fluctuation is known to represent profit fluctuation by Brownian motion with drift:

$$d\eta_t = \mu dt + \sigma dW_t$$

When profit fluctuations are modeled as Brownian motion with drift, the value of corporate assets is

$$V_T^H = \eta_T\left(-\frac{1}{\rho}e^{-\rho(H-T)} + \frac{1}{\rho}\right) + \mu\left(-\frac{He^{-\rho(H-T)}}{\rho} - \frac{e^{-\rho(H-T)}}{\rho^2} + \frac{1}{\rho^2}\right)$$

In the case of $H \to \infty$,

$$V_T^\infty := \lim_{H\to\infty} V_T^H = \frac{\eta_T}{\rho} + \frac{\mu}{\rho^2}$$

Assuming that the level of corporate asset value in insolvency is the amount of debt, the level of profits in insolvency $\eta^{D,H}$ satisfies following equation:

$$D = \eta^{D,H}\left(-\frac{1}{\rho}e^{-\rho(H-T)} + \frac{1}{\rho}\right) + \mu\left(-\frac{He^{-\rho(H-T)}}{\rho} - \frac{e^{-\rho(H-T)}}{\rho^2} + \frac{1}{\rho^2}\right).$$

In the case of $H \to \infty$, the solution is

$$\eta^{D,\infty} = \rho D$$

Therefore, the probability of insolvency is

$$P(V_T^H < D) = P(\eta_T < \eta^{D,H}) = \Phi(-DD_E^H) \tag{6}$$

where

$$DD_E^H := \frac{(\eta_0 + \mu T) - \eta^{D,H}}{\sigma\sqrt{T}}. \tag{7}$$

The last equality in Eq. (6) follows from the fact that the profit η_T has normal distribution with mean $\eta_0 + \mu T$ and variance $\sigma^2 T$.

Furthermore, consider a model in which profit is expressed as the difference between sales and expenses. That is, let s_t be the instantaneous sales at time t and c_t be the instantaneous expenses. The instantaneous profit is represented by the difference

$$\eta_t = s_t - c_t$$

A model of profit can be derived from the specific model of instantaneous sales and expenses. Let us consider the case where instantaneous sales are represented by a Brownian motion with drift:

$$ds_t = \mu^S dt + \sigma^S dW_t^S$$

Assume further that the cost is modeled as an affine function of sales, assuming an order-taking firm.

$$c_t = a_0 + a_1 s_t$$

where a_0 is the instantaneous fixed cost and a_1 is a parameter representing the variable cost ratio. In this case, the instantaneous profit of the firm is

$$\begin{aligned} d\eta_t &= \mu dt + \sigma dW_t^S, \quad \eta_0 = s_0 - c_0 = (1 - a_1)s_0 - a_0 \\ \mu &= (1 - a_1)\mu^S, \ \sigma = (1 - a_1)\sigma^S \end{aligned}$$

Parameter values $\mu = (1 - a_1)\mu^S$ and $\sigma = (1 - a_1)\sigma^S$ are obtained by estimating the sales model using sales information. Then, we are able to estimate the default distance DD_E^H with observed debt value D.

4 Classification method: LSPC and MT-LSPC

We employ LSPC [4] and Multi-task LSPC [3] to estimate class-posterior probability $p(y|\boldsymbol{x}, t)$. The structure of LSPC is a linear combination of kernel functions, and its parameters are learned through the least-squares fitting of the true class' posterior probability. Because the process of training the LSPC is implemented by solving a regularized system of linear equations, the LSPC is computationally efficient and numerically stable. In this section, we briefly describe LSPC and Multi-task LSPC.

A brief description of the LSPC is as follows. Recall that $X \subset \mathbb{R}^p$ be a domain of the inputs and $Y = \{0, \ 1\}$ be the set of classes representing non-default and default. Consider a joint probability density of $p(\boldsymbol{x}, \ y)$ on $X \times Y$. Given N independent and identically distributed paired samples of input $\boldsymbol{x}$ and output y, we estimate the class-posterior probability $p(y|\boldsymbol{x})$. In the estimation algorithm of the LSPC, we model the class-posterior probability $p(y|\boldsymbol{x})$ using the linear model, as follows:

$$p(y|\boldsymbol{x}) = \alpha_{y,1} k(\boldsymbol{x}, \boldsymbol{x}_1) + \alpha_{y,2} k(\boldsymbol{x}, \boldsymbol{x}_2) + \cdots + \alpha_{y,N} k(\boldsymbol{x}, \boldsymbol{x}_N)$$

$$= \boldsymbol{\alpha}_y^\top \phi(\boldsymbol{x}),$$

where $\boldsymbol{\alpha}_y = (\alpha_{y,1},\ \alpha_{y,2},\ \ldots, \alpha_{y,N})^\top \in \mathbb{R}^N$ is the N-dimensional parameter vector for class y, $k(\boldsymbol{x}, \boldsymbol{x}')$ represent the kernel functions and $\phi(\boldsymbol{x}) = (k(\boldsymbol{x}, \boldsymbol{x}_1),\ k(\boldsymbol{x}, \boldsymbol{x}_2),\ \ldots, k(\boldsymbol{x}, \boldsymbol{x}_N))^\top$. In this study, we employ the Gaussian kernel

$$k(\boldsymbol{x}, \boldsymbol{x}') = \exp\left(-\frac{\|\boldsymbol{x} - \boldsymbol{x}'\|^2}{\sigma^2}\right)$$

for kernel functions. To estimate the parameter $\boldsymbol{\alpha}_y = (\alpha_{y,1}, \alpha_{y,2}, \ldots, \alpha_{y,N})^\top$, we solve the following optimization problem with regularization term

$$\hat{\boldsymbol{\alpha}}_y = \underset{\boldsymbol{\alpha}_y}{\operatorname{argmin}} \frac{1}{2N} \sum_{n=1}^{N} \boldsymbol{\alpha}_y^\top \phi(\boldsymbol{x}_n) \phi(\boldsymbol{x}_n)^\top \boldsymbol{\alpha}_y - \frac{1}{N} \sum_{n=1}^{N} \boldsymbol{\alpha}_y^\top \phi(\boldsymbol{x}_n) + \frac{\lambda}{2} \|\boldsymbol{\alpha}_y\|^2,$$

The solution to the optimization problem is given as follows (see [3] for details):

$$\hat{\boldsymbol{\alpha}}_y = \left(\boldsymbol{\Phi}^\top \boldsymbol{\Phi} + \lambda N \boldsymbol{I}_N\right)^{-1} \boldsymbol{\Phi}^\top z_y^\top$$

where

$$\boldsymbol{\Phi} = (\phi(\boldsymbol{x}_1),\ \phi(\boldsymbol{x}_2), \ldots, \phi(\boldsymbol{x}_N),)^\top$$

and

$$z_y = \left(\delta_{y,y_1},\ \delta_{y,y_2}, \ldots, \delta_{y,y_N}\right)^\top.$$

To ensure that the estimated class-posterior probability is a non-negative value under the estimated $\hat{\boldsymbol{\alpha}}_y$, we round up class-posterior probability as follows:

$$\hat{p}(y|\boldsymbol{x}) = \frac{\max(0,\ \hat{\boldsymbol{\alpha}}_y^\top \phi(\boldsymbol{x}))}{\sum_{\tilde{y} \in \{0,1\}} \max(0,\ \boldsymbol{\alpha}_{\tilde{y}}^\top \phi(\boldsymbol{x}))}.$$

Multi-task LSPC is an extended version of LSPC, where posterior probability $p(y|\boldsymbol{x})$ is modeled by

$$p(y|\boldsymbol{x}) = (\boldsymbol{\beta}_{y,0} + \boldsymbol{\beta}_{y,t})^t \phi(\boldsymbol{x})$$

Here, $t \in \{1, 2, \ldots, T\}$ is the task index, which represents the industrial sector index in our analysis, $\boldsymbol{\beta}_{y,0}$ is the common parameter for all tasks, and $\boldsymbol{\beta}_{y,t}$ is individual parameter for task t. To estimate the parameters, we solve the following optimization

problem with regularization term

$$\hat{\boldsymbol{\beta}}_y = \underset{\boldsymbol{\beta}_y}{\operatorname{argmin}} \frac{1}{2N} \sum_{n=1}^{N} (\boldsymbol{\beta}_{y,0} + \boldsymbol{\beta}_{y,t_n})_y^\top \phi(\boldsymbol{x}_n)\phi(\boldsymbol{x}_n)^\top (\boldsymbol{\beta}_{y,0} + \boldsymbol{\beta}_{y,t_n})$$
$$- \frac{1}{N} \sum_{n=1}^{N} (\boldsymbol{\beta}_{y,0} + \boldsymbol{\beta}_{y,t_n})_y^\top \phi(\boldsymbol{x}_n) + \frac{\lambda}{2} \|\boldsymbol{\beta}_{y,0}\|^2 + \frac{\gamma}{2T} \sum_{t=1}^{T} \|\boldsymbol{\beta}_{y,t}\|^2.$$

Under the reformulation provided in [3], class posterior probability is obtained as follows:

$$\hat{p}(y|\boldsymbol{x}) = \frac{\max(0, \hat{\boldsymbol{\mu}}_y^\top \boldsymbol{\xi}(\boldsymbol{x}, t))}{\max(0, \hat{\boldsymbol{\mu}}_0^\top \boldsymbol{\xi}(\boldsymbol{x}, t)) + \max(0, \hat{\boldsymbol{\mu}}_1^\top \boldsymbol{\xi}(\boldsymbol{x}, t))}.$$

where

$$\boldsymbol{\xi}(\boldsymbol{x}, t) = \left(\psi(\boldsymbol{x}_1, t_1)^\top \psi(\boldsymbol{x}, t), \ldots, \psi(\boldsymbol{x}_N, t_N)^\top \psi(\boldsymbol{x}, t) \right)^\top,$$
$$\hat{\boldsymbol{\mu}}_y^\top = \left(\boldsymbol{\Psi}\boldsymbol{\Psi}^t + \frac{\gamma N}{T} I_N \right)^{-1} \boldsymbol{z}_y,$$
$$\boldsymbol{\Psi} = (\psi(\boldsymbol{x}_1, t_1), \ldots, \psi(\boldsymbol{x}_N, t_N))^\top,$$

Here, $\psi(\boldsymbol{x}_1, t_1)$ is a vector whose first N elements are $\sqrt{\gamma/(T\lambda)}\phi(\boldsymbol{x})^\top$, and N elements from the $(Nt+1)$-th element to the $(Nt+N)$-th element are equals to $\phi(\boldsymbol{x})$, and other elements are 0.

5 Empirical analysis

5.1 Original data

We use a dataset consisting of Japanese unlisted firms in the construction industry (Sector 1), manufacturing industries (Sector 2) and wholesale and retail industry (Sector 3) in 2011 and 2012. The close nature of the training and test data is desirable for our analysis. There were no significant changes in the economic situation in Japan between 2011 and 2012. For this reason, these period were chosen as the training and test data period. The samples are taken from Teikoku Data Bank, Ltd.'s database, which consists of information gathered through a company survey conducted by the company. Each sample has an identification key consisting of a company code and year. Each sample is labeled as a bankrupt firm ($y_i = 1$) if a firm goes bankrupt within one year of the end of the key year. If a firm is not bankrupt one year after the end of the key year, the sample is labeled as a non-bankrupt firm ($y_i = 0$). The original samples include sales and net income for the past three years including the key year. The sales and net income are yearly observed. In addition, total assets, liabilities, sales to the three main customers, amounts paid to the three main suppliers, and the

Table 1 Sample size before under-sampling

	Sec 1		Sec 2		Sec 3	
	$y = 0$	$y = 1$	$y = 0$	$y = 1$	$y = 0$	$y = 1$
Training data (2011)	27,189	90	17,793	28	23,433	42
Test data (2012)	26,971	57	18,160	39	23,861	31

score of each main supplier observed in the key year are also available. The features CTS_i^{in}, CTS_i^{out}, HHI_i^{in}, HHI_i^{out} are calculated using these observed values. In addition, we include the average of sales ($AveSales$) and the standard deviation of sales ($StdSales$).

5.2 Calculating distance to default

We calculate the values of distance to default DD_E^H for each sample using three years of sales. To calculate Distance to Default DD_E^H, we need the estimated values of sales parameter μ^S, σ^S, s_0 and parameters a_0, a_1 which relates to calculating costs. Since the observed sales and net incomes are time-series data at only 3 time points, it is hard to estimate the three parameters μ^S, σ^S, s_0. Thus, we set $\mu^S = 0$, and estimated the σ^S by sample standard deviation, estimated s_0 by sample average of observed sales. We set the value of the variability cost ratio to $a_1 = 0.7$. This value is approximately the level of small firms in the Japanese non-manufacturing industry. Fixed cost a_0 is calculated as the difference between observed profits and $a_1 s_0$. We set $T = 1$ and time-horizon H is set to $H = 20$ or $H = \infty$. Accordingly, we obtain two kinds of values, DD_E^{20} and DD_E^{∞}, for distance to default. For estimating default levels of profits η^D, we set the risk-adjusted discount rate ρ to an average interest rate that the sample firm pays.

5.3 Preprocessing

In data preprocessing steps, we employ min-max scaling, which rescales the range of features to the range in [0, 1], to mitigate the size effect of the variables. Then, we split all data into training and test data. Specifically, we use the samples whose key year is 2011 for training data, and we use samples whose key year is 2012 for test data.

Our original dataset is imbalanced data, that is the number of records of bankruptcy corporate is only a few. Thus, we applied under-sampling in order to balance the number of bankrupt records and non-bankrupt records. Specifically, we randomly under-sampled the non-bankrupt firms so that the number of firms in the non-bankrupt firms matched the number of firms in the bankrupt firms. In this way, we created 100 sub-datasets with a balanced number of records for non-bankrupt and bankrupt firms, and performed the analysis on them. For each sub-datasets from training data, we performed 3-fold cross-validation for estimating values of the hyper-parameters λ and γ. Table 1 shows the size of samples before conducting under-sampling.

5.4 Performance metrics

To evaluate the accuracy of the classification results, we employ a widely used measure called the accuracy ratio (ACC). The ACC measures the overall effectiveness of the model. The ACC is defined as the number of correct predictions over the number of predictions. It is defined as follows:

$$\mathrm{ACC} = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP represents the number of true positive predictions, TN represents the number of true negative predictions, FP represents the number of false positive predictions, and FN represents the number of false negative predictions. A higher ACC implies improved prediction performance, and the ACC of the random prediction is 0.5.

5.5 Results on efficiency of the single-features

In this subsection, we verify whether each single feature is effective for bankruptcy prediction. That is, we perform bankruptcy prediction using only one feature and test whether the result is superior to a random prediction. The results of the prediction are measured by the ACC As the value of the ACC for random forecasts is 0.5, and we focus on whether the ACC for single-indicator forecasts exceeds 0.5. We do not use LSPC here, but logistic regression:

$$p(y|x) = \frac{1}{1 + \exp(-(\beta_0 + \beta_1 x))}$$

where x is a single feature. By using logistic regression, the relationship between the direction of increase or decrease in the value of a feature and the direction of increase or decrease in bankruptcy probability can be confirmed.

Table 2 presents the performance of logistic regression using single features.

Several values listed in Table 2 are above 0.5, suggesting that these proposed indicators include information related to bankruptcy. In particular, Table 2 shows that DD_E tended to be effective in all industries. Creditworthiness of customer CTS_i^{in} are totally effective than CTS_i^{out}. The efficiency of HHI differs depending on the industry. For example, in the construction industry (Sec 2), the diversification of suppliers is effective. On the other hand, the diversification of customers affects bankruptcy for the manufacturing industry. For the retail and wholesale industry, both clients' and suppliers' diversification are effective.

In addition, we checked the values of the coefficients β_1 to clarify the relationship between feature values and bankruptcy occurrence. Table 3 shows the average of coefficients of each feature in single-feature logistic regression.

The coefficient of CTS tends to be negative, suggesting that the higher the creditworthiness of customer and supplier firms, the less likely they are to go bankrupt. The coefficient of HHI tends to be negative. This result indicates that the higher the dispersion of transaction shares, the lower the likelihood of bankruptcy. These results

Table 2 ACC for single feature model

Feature	ACC			
	All	Sec 1	Sec 2	Sec 3
CTS_i^{in}	0.526	0.491	0.519	0.537
CTS_i^{out}	0.489	0.495	0.505	0.535
HHI_i^{in}	0.524	0.510	0.488	0.509
HHI_i^{out}	0.533*	0.488	0.561	0.526
$AveSales$	0.492	0.517	0.464	0.578*
$StdSales$	0.495	0.511	0.477	0.503
DD_E^{∞}	0.583*	0.608*	0.595*	0.527
DD_E^{20}	0.562*	0.539	0.607*	0.435

The values are the average of results from 100 sets of sub-samples. A symbol * is indicated for results with an ACC value significantly greater than 0.5. The significance level is 10%, which is the situation where the value of 0.5 is below the 1.28σ range from the sample mean of ACC

Table 3 Estimated coefficients of single-feature logistic regression

	All	Sec 1	Sec 2	Sec 3
CTS_i^{in}	−0.13	−0.19	−0.25	0.26
CTS_i^{out}	−0.43	−0.15	−0.14	−0.41
HHI_i^{in}	−0.23	−0.20	−0.32	0.21
HHI_i^{out}	−0.33	−0.43	0.03	−0.15
$AveSales$	−0.06	−0.29	0.20	−0.64*
$StdSales$	0.17	−0.02	0.28*	−0.25
DD_E^{∞}	−0.51*	−0.30*	−0.24*	−0.26
DD_E^{20}	−0.02	−0.22*	0.00	−0.03

A symbol * is indicated for results with a coefficient value significantly not equal to 0. The significance level is 10%, which is the situation where the value of 0 is out of the 1.28σ range from the sample mean of coefficients

are consistent with the results in [7, 8], which explain the relationship between the degree of counterparty diversification, the creditworthiness of counterparties, and the probability of default with a structural approach. The coefficient of the default distance takes a negative value. This result implies that the larger the default distance, the lower the likelihood of bankruptcy. This result is reasonable considering the meaning of default distance.

5.6 Default prediction by MT-LSPC

In this subsection, we verify the effectiveness of default prediction via multi-task learning. The feature sets used for our prediction are Set 1 and Set 2. Set 1 consists of

Table 4 Accuracy obtained for the test data

		Sec 1	Sec 2	Sec 3
Feature set 1	LSPC	0.522	0.491	**0.522**
	MT-LSPC	**0.533**	**0.498**	0.517
Feature set 2	LSPC	0.530	0.526	**0.534**
	MT-LSPC	**0.554**	**0.544**	0.528

The accuracies of LSPC and MT-LSPC are compared, with the higher accuracy figures in bold

the following features:

$$\{CTS_i^{in},\ CTS_i^{out},\ HHI_i^{in}, HHI_i^{out}, AveSales, StdSales\}$$

Set 2 consists of the following features:

$$\{CTS_i^{in},\ CTS_i^{out},\ HHI_i^{in}, HHI_i^{out}, AveSales, StdSales,\ DD_E^{\infty},\ DD_E^{20}\}$$

Set 1 does not include distance to defaults. Therefore, Set 1 consists only of features that do not use observed debt. Set 2 is the addition of distance to default to Set 1.

Table 4 shows the ACC for single-task LSPC and MT-LSPC with feature set 1 and set 2.

Most of the ACC values obtained were above 0.5. This result indicates that the features employed in this study are effective when compared to random prediction, and further suggests that the prediction methods, LSPC and MT-LSPC, are effective in predicting bankruptcy. Comparing the ACC of the MT-LSPC with the ACC of the single-task LSPC, we can see that the MT-LSPC is more effective, except in Sec 3.

These results indicate that the features proposed in this study are effective as input information for bankruptcy prediction. In addition, it was confirmed that the prediction accuracy is improved by multi-task learning.

6 Conclusion

We proposed a bankruptcy prediction method using transaction-based corporate features and features called "distance to default" which are derived from the structural approach of credit risk assessment. Instead of the traditional single-task type of default prediction model estimation, in which models were constructed independently for each industry, we proposed a model estimation by multi-task learning, in which models for multiple industries are constructed simultaneously. We used the Japanese corporate dataset to test the effectiveness of the proposed features and classification power of the MT-LSPC. The results reported in the empirical study show that using new features is superior to predicting bankruptcy randomly. In particular, the "distance to default" tends to be relatively useful among proposed features. Moreover, for bankruptcy prediction for several industries, the forecasting accuracies of MT-LSPC are found to

exceed that of LSPC. These results suggest that it is possible to construct prediction models by multi-task learning even with small-sized samples.

The method proposed in this study may enable new bankruptcy prediction for companies that have not prepared financial statements, which the conventional prediction methods were not able to apply. However, the accuracy of the proposed method seems insufficient to use in actual financial operations. It is expected to propose additional features to realize sufficient accuracy. Some of the features that were suggested to be useful in this analysis, such as the degree of diversification of business partners, can be interpreted as features derived from the business-to-business transaction network. In future work, comprehensive validation of features derived from business-to-business transaction networks is expected. It is also desirable to use features derived from textual data of company survey reports as input information for the forecast.

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Data availability As the data used in this analysis is not open data, we have omitted any description of data availability.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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